



YELLOWSENSE TECHNOLOGIES

AI-Powered Cargo Forecasting & Trade Incentive Intelligence Platform

Secure Maritime Intelligence Infrastructure for Indian Ports

Empowering port authorities with predictive analytics, trade intelligence and AI-driven decision support to optimize operations, boost revenue and strengthen India's maritime competitiveness.



Predict what's arriving

AI models forecast cargo, vessels & demand with high accuracy.



Optimize port operations

Better planning for berths, yard, workforce & resources.



Maximize revenue

Data-driven incentive recommendations to attract more trade.



Secure & compliant

Confidential computing ensures privacy, security & DPDP compliance.



Intelligence at scale

Unified platform for real-time insights across ports, agencies & stakeholders.

Smarter Decisions. Stronger Ports. Greater India.

- ✓ AI & Predictive Analytics
- ✓ Trade & Incentive Intelligence
- ✓ Digital Twin & Scenario Simulation
- ✓ Secure Data Collaboration
- ✓ Actionable Insights. Real Impact.



Vision:

To build India's next-generation AI infrastructure for maritime forecasting and trade intelligence.



Mission:

Deliver predictive insights and AI-powered decisions for efficient, profitable & future-ready ports.



Impact:

Lower congestion. Higher efficiency.
Maximized trade. Stronger economy.





The Challenges

Legacy systems and fragmented data prevent ports from seeing ahead, acting early, and staying competitive.



Challenge 1

Historical Forecasting



Most ports rely on:

- Previous year statistics
- Quarterly reports
- Manual spreadsheets

✗ These approaches fail when:

- Trade wars occur
- Fuel prices fluctuate
- Geopolitical conflicts arise
- Shipping routes change



Challenge 2

No Global Visibility



Current systems cannot answer:

- Which vessels are heading towards India?
- What commodities are they carrying?
- Which trade lanes are growing?
- Which cargo categories are declining?



Challenge 3

Lack of Trade Intelligence



Ports know:

"What arrived."

But they do not know:

"What is about to arrive."



Challenge 4

Reactive Incentives



Trade incentives are usually introduced after cargo decline has already occurred.



The Result

Delayed decisions lead to real and measurable losses.



Lost Revenue

Missed opportunities and reduced earnings.



Congestion

Inefficient planning causes delays and bottlenecks.



Reduced Competitiveness

Higher turnaround times and lower global rankings.



Our Solution

Catered to Cochin Port and others

An AI-powered, multi-agent intelligence platform that transforms uncertain data into actionable decisions for smarter port operations and trade growth.



Predictive Cargo Analytics

See What's Coming. Plan Ahead.

AI models predict vessel arrivals, cargo volumes, and trade patterns with high accuracy.

- ✓ Vessel ETA & volume forecasting
- ✓ Commodity & route demand prediction
- ✓ Anomaly detection & risk alerts



Trade Incentive Intelligence

Maximize Benefits. Drive Trade.

Identify policy leakages and uncover optimal incentives to boost cargo movement.

- ✓ Policy & incentive tracking
- ✓ Leakage detection & impact analysis
- ✓ Incentive utilization recommendations



Secure Data Collaboration

Share Securely. Decide Confidently.

Confidential computing enables secure, privacy-preserving data sharing.

- ✓ Multi-agency data collaboration
- ✓ Privacy-preserving analytics
- ✓ Role-based access & audit trails



Powered by Multi-Agent Framework



Data Ingestion Agent

Collects data from multiple sources in real-time



Processing Agent

Cleans, standardizes & enriches data



Analytics Agent

Runs AI/ML models for forecasts & insights



Insight Agent

Generates recommendations & actionable insights



Decision Support Agent

Supports authorities with data-driven decisions



Improved Forecast Accuracy

Up to 85% improvement in cargo demand prediction



Reduced Vessel Wait Time

Better planning for berths & resources



Higher Incentive Realization

Maximize policy benefits & revenue



Secure & Compliant

Built on confidential computing & DPDP compliance



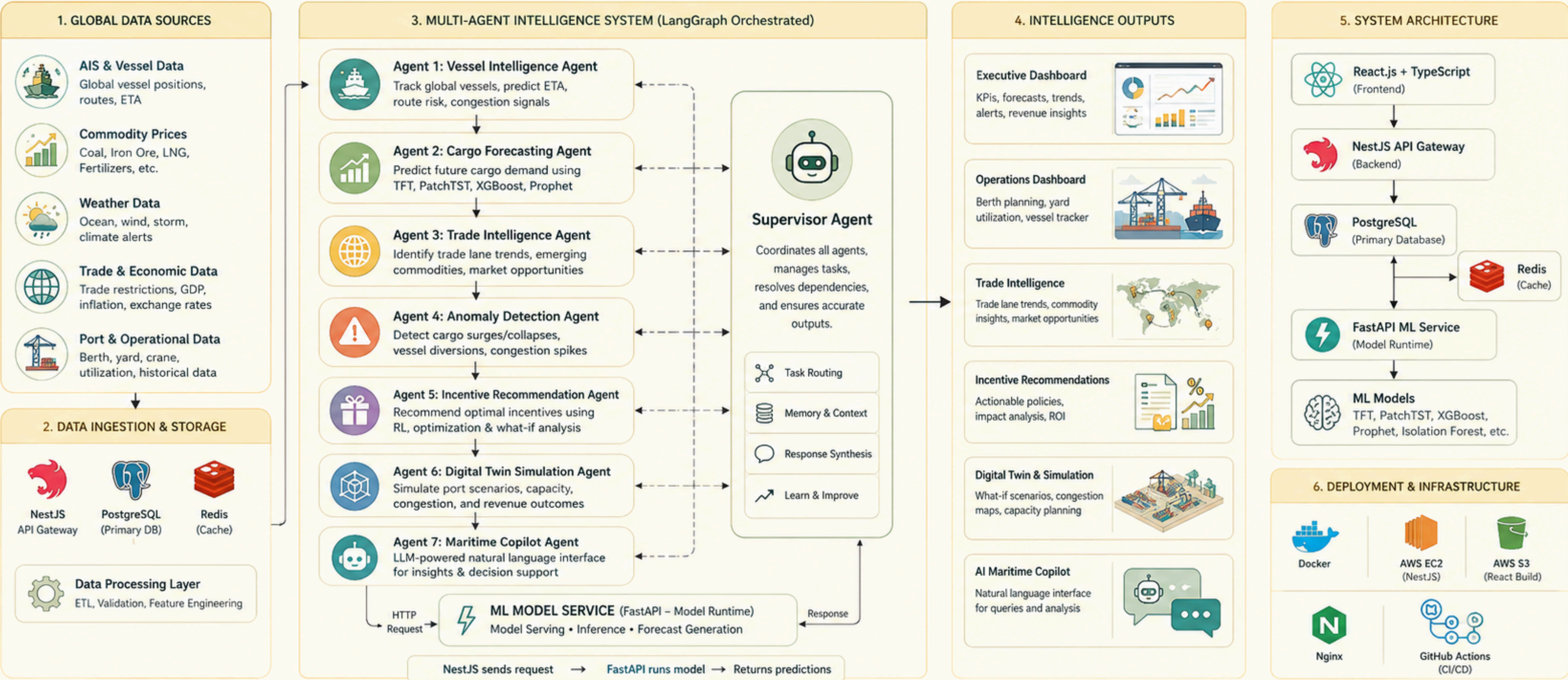
Stronger Collaboration

Unified platform for ports, govt. & trade stakeholders



AI-Powered Cargo Forecasting & Trade Intelligence Platform

Multi-Agent Intelligence Pipeline Overview



Proactive Decision Making
Predict before cargo arrives.



Improved Efficiency
Optimize resources & reduce congestion.



Data Security & Privacy
Confidential computing ready architecture.



Higher Revenue
Smarter incentives, better utilization.



YELLOWSENSE TECHNOLOGIES

Marine Impact Areas

Intelligent Insights. Smarter Ports. Stronger Maritime Ecosystem.



Cargo Predictability

Improve forecasting accuracy for better planning and resource allocation.

↑ 25–35%
Forecast Accuracy



Vessel Wait Time

Reduce congestion and accelerate turnaround at berths.

↓ 20–30%
Average Wait Time



Operational Efficiency

Streamline inter-agency coordination and port operations.

↑ 15–25%
Operational Efficiency



Revenue Enhancement

Optimize incentives and maximize cargo throughput & port revenues.

↑ 10–20%
Revenue Uplift



Risk & Disruption Reduction

Detect anomalies early and mitigate disruptions proactively.

↓ 30–40%
Disruption Events



Sustainability Impact

Optimize routes and reduce idle time for a greener maritime ecosystem.

↓ 15–20%
Emission Reduction



Stakeholder Collaboration

Enable secure data sharing and strengthen inter-agency collaboration.

100%
Secure & Compliant



Stronger Ports. Stronger India.

Data-driven decisions for a resilient, efficient and future-ready maritime ecosystem.



Predict Better. Operate Smarter. Grow Together.

Building India's Intelligent Maritime Future.



Team Strength

A passionate team of builders, researchers & domain experts driving India's Maritime Intelligence Revolution.



15+

Team Members



50+

Years Combined Experience



Deep-Tech

AI, Data, Cloud & Maritime Expertise



One Mission

Smarter Ports. Stronger India.

A COMPLETE TEAM. EVERY EXPERTISE. ONE VISION.



AI/ML & Data Science

Building intelligent models for forecasting, anomaly detection & optimization.



Data Engineering & Platform

Robust pipelines, scalable infrastructure & real-time data processing.



Product & Engineering

Designing intuitive platforms and building high-performance systems.



Security & Confidential Computing

Ensuring data privacy, DPDP compliance & secure collaboration.



Maritime Domain Experts

Deep understanding of ports, vessels, trade, operations & industry dynamics.



Business & Strategy

Driving partnerships, growth, market outreach & long-term impact.



All the strengths required to build, scale and deliver India's most advanced maritime intelligence platform.

OUR EDGE



Domain + Deep Tech

Unique blend of maritime domain expertise and advanced AI capabilities.



Security by Design

Confidential computing & DPDP-compliant architecture from the ground up.



Problem-First Approach

Solving real problems of ports, government & trade ecosystem.



Collaborative Spirit

Strong culture of innovation, integrity and impact.

OUR STRENGTHS



AI-First Mindset

Building intelligent systems that learn and adapt.



Maritime Domain Expertise

Deep understanding of ports, vessels, trade & operations.



End-to-End Capability

From data ingestion to insights, recommendations & action.



Secure & Scalable

Built for scale with security, resilience & reliability.



Impact Driven

Focused on measurable value & national impact.



Diverse Minds. Deep Expertise. One Mission.

Powering Smarter Ports. Enabling a Stronger India.





Market Traction / Market Opportunity

A \$4.6B+ opportunity in the global maritime intelligence market



\$4.6B+

Market Size (2024)

Global Maritime Analytics
Market



14.8%

CAGR (2024–2030)

Strong growth driven by AI,
digitalization & trade growth



\$15B+

Addressable Market

Ports, Shipping Lines, Govt.
Agencies & Logistics Players



₹3,000 Cr+

India Opportunity

Maritime analytics & digital
transformation by 2030



KEY INDUSTRY DRIVERS



Surging Global Trade
Growing cargo volumes
and complex supply
chains



AI & Data Revolution
Real-time intelligence
unlocking predictive
decision making



Policy & Compliance
Data governance,
security & sustainability
mandates



Port Modernization
Digital ports & smart
infrastructure driving
demand



Geopolitical Shifts
Trade realignment
creating new route
opportunities

WHO WE SERVE



Port Authorities



Shipping Lines



Logistics &
Terminal Operators



Government
Agencies



Trade &
Policy Bodies

OUR COMPETITIVE ADVANTAGE



AI-First Platform
Advanced forecasting,
anomaly detection &
prescriptive insights



Confidential & Secure
Confidential computing
ensuring privacy,
compliance & trust



End-to-End Solution
From data ingestion to
decision support in a
single platform



Ecosystem Approach
Built for collaboration
across ports, agencies
and stakeholders



Proven Use-Cases
Pilot-ready solutions
delivering measurable
business impact

TRACTION SNAPSHOT

- ✓ Pilot engagement with leading port authorities
- ✓ Strong partnerships with academic & research institutions
- ✓ Secured recognition in national innovation challenges.
- ✓ Growing pipeline of enterprise & government use cases
- ✓ Positive feedback from domain experts & early users



Massive Market. Strong Tailwinds. The Right Time.

Positioned to lead India's Maritime Intelligence Revolution.





External Funding

Seeking partnership to scale India's Maritime Intelligence Infrastructure



FUNDING ASK



₹98 Lakhs

To accelerate product development, pilot deployments, and scale our AI-powered maritime intelligence platform.

FUND UTILIZATION



- **40%**
AI R&D & Model Development
- **20%**
Pilot Deployments & Partnerships
- **15%**
Product Engineering & Platform Scaling
- **15%**
Sales, Marketing & Business Development
- **10%**
Operations & Compliance

WHAT THIS FUNDING ENABLES



Advanced AI Development

Build next-gen forecasting, anomaly detection & digital twin capabilities



Pilot to Scale

Expand pilots to multiple ports and government agencies across India



Platform Scaling

Strengthen infrastructure, security, and data pipelines for national-scale deployment



Stronger Partnerships

Deepen collaborations with ports, ministries and ecosystem partners

WHY INVEST IN YELLOWSENSE?



Large & Growing Market

\$4.6B+ global opportunity with strong CAGR of 14.8%



First-Mover Advantage

India's first AI + Confidential Computing maritime platform



Proven Traction

Pilot projects, strategic partnerships & strong domain validation



High Impact

Solving real problems for ports, government & trade ecosystem

POTENTIAL FUNDING PARTNERS



Government Grants & Innovation Funds



VCs & Angel Investors



Corporate Strategic Investors



Maritime & Logistics Industry Partners



Research & Academic Collaborations



Your partnership will help us build India's most advanced maritime intelligence infrastructure and create long-term value for the nation.



Invest in Intelligence. Power India's Ports. Shape the Future of Maritime Trade.

Together, we build smarter ports for a stronger India.





Thank You!

Smarter Data. Stronger Ports. Greater India.



Our Startup Vision

To build the **intelligence layer** for India's maritime ecosystem by transforming fragmented data into predictive insights, enabling ports and stakeholders to make smarter decisions, unlock growth, and power a future-ready maritime India.



Data to Decisions

Turn complex, disparate data into accurate forecasts and actionable intelligence.



Ecosystem Impact

Empower ports, shipping lines, agencies, and policy makers to collaborate and grow together.



Sustainable Growth

Drive efficient operations, reduce costs, and promote a sustainable maritime ecosystem.



Nation First

Strengthen India's global maritime position and contribute to economic progress.



Future Ready

Build resilient, adaptive and AI-powered infrastructure for the maritime future.



"Our mission is to empower every port and every decision with intelligence — for a **stronger, smarter, maritime India.**"



Together, let's build
Intelligent Ports. Resilient India.

